



Critical Systems Labs
Innovating Safely

From Formal Proof to Certification Evidence: The Role of Assurance Arguments

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September 3, 2026



VANCOUVER 2026

Abstract

Formal verification can provide particularly strong evidence in support of certification claims. However, using formal verification results effectively in a certification context can be challenging, particularly when their scope, assumptions, and relevance to broader system-level claims need to be understood and communicated. This presentation identifies some of these challenges and examines what is needed for formal verification results to become compelling certification evidence.

This presentation explores how structured assurance arguments can provide a framework for addressing these challenges. Such arguments are increasingly used for complex systems, and in some industries are becoming an expected part of the certification package. These arguments can make explicit how formal verification contributes to higher-level claims, how it relates to other forms of evidence, and how together these elements support confidence in the overall safety case.

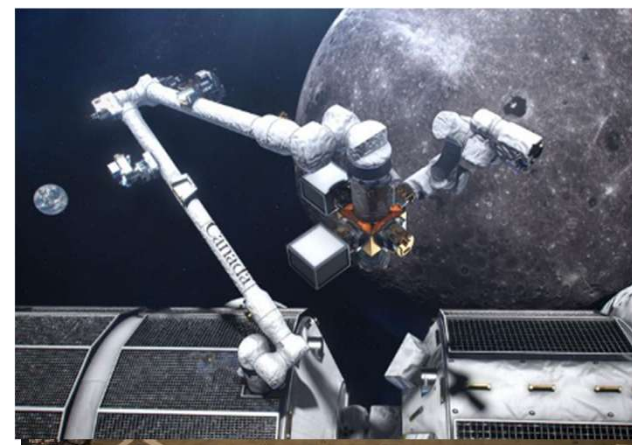
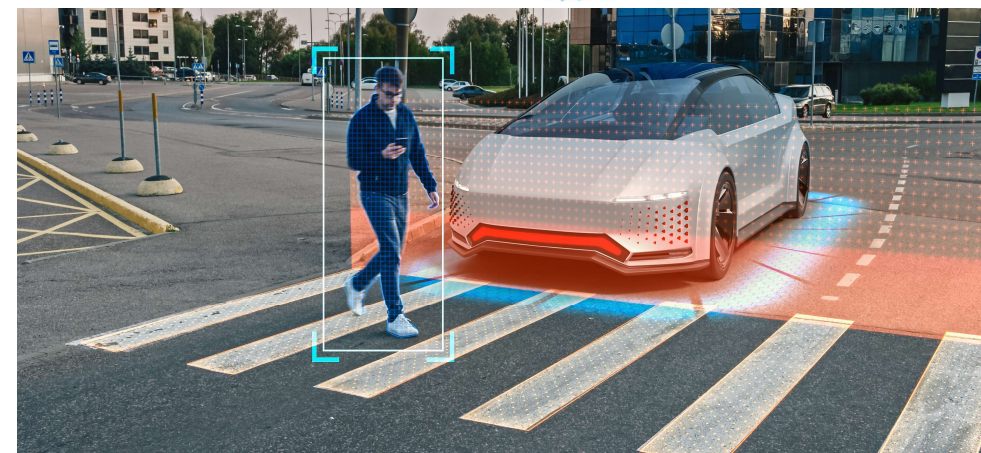
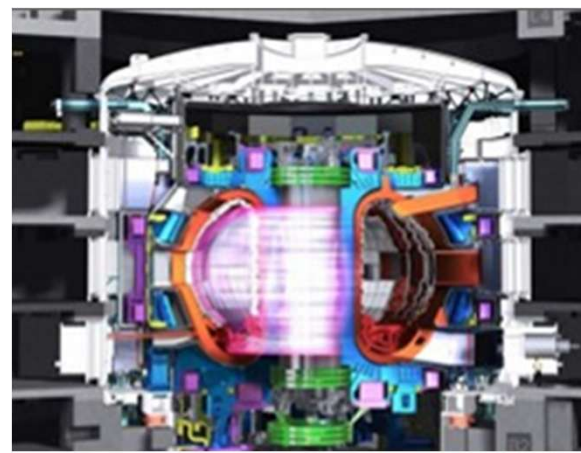


Dr. Laure Millet is the Vice President Research at Critical Systems Labs Inc. (CSL), with an international reputation for expertise in system safety for complex systems.

Most of her projects address safety challenges associated with emerging technologies, particularly AI and ML, as well as their use to support and improve system safety. Laure specializes in systems engineering and safety engineering methodologies for compliance with

standards such as ISO 26262, ISO 21448 (SOTIF), and ISO/PAS 8800. She also contributes to the development of standards, particularly sections addressing systems that incorporate AI or are developed using AI.

Laure is also a co-lead of CSL's product development team of highly skilled researchers and developers for the Socrates Assurance Case Editor. She earned a Ph.D. in Computer Science, with a focus on formal methods, from Pierre and Marie Curie Université (France).



A proof is evidence... but...

Formal verification shows:

- a stated property holds;
- for a specific model; under defined assumptions.

Certification still asks:

- What system claim does it support?
- Is the model adequate?
- What remains outside the proof?

FM Experts



Certification Experts





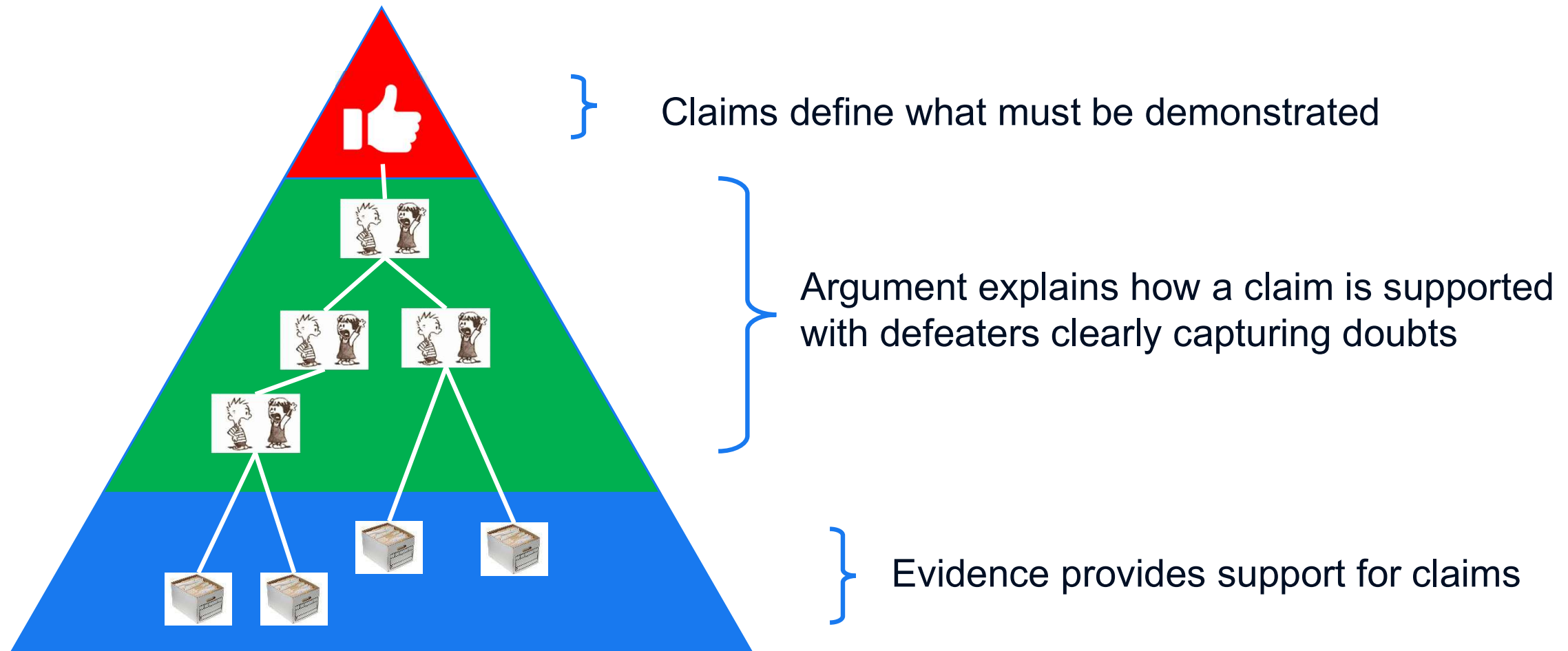
- What exactly was proved?
- Is the model representative?
- What is outside the scope?
- Is the toolchain qualified?
- Can the result be independently reviewed?
- Is the traceability to the system claim complete?
- What do we do if the system change?

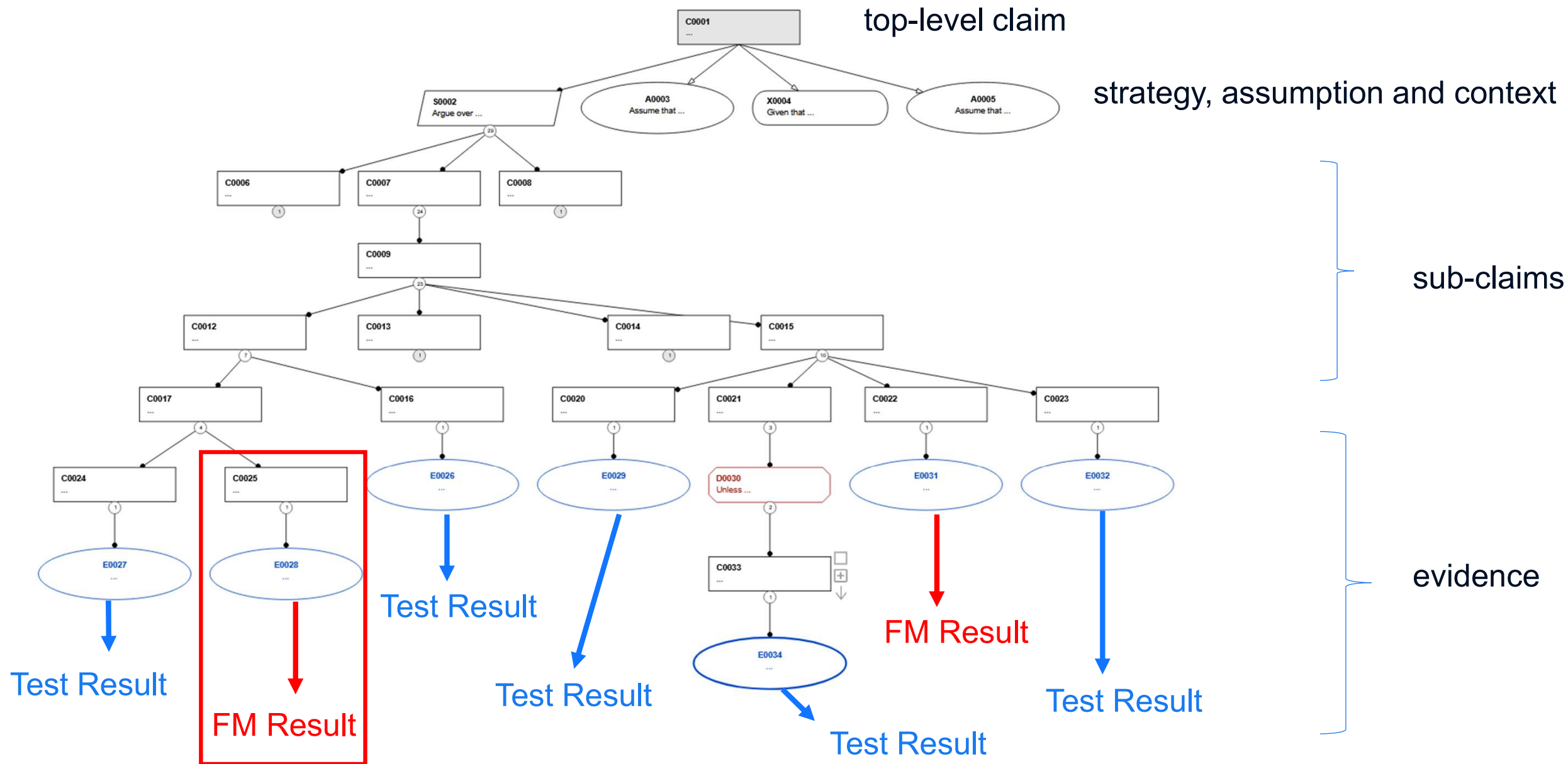
From Problem to Solution

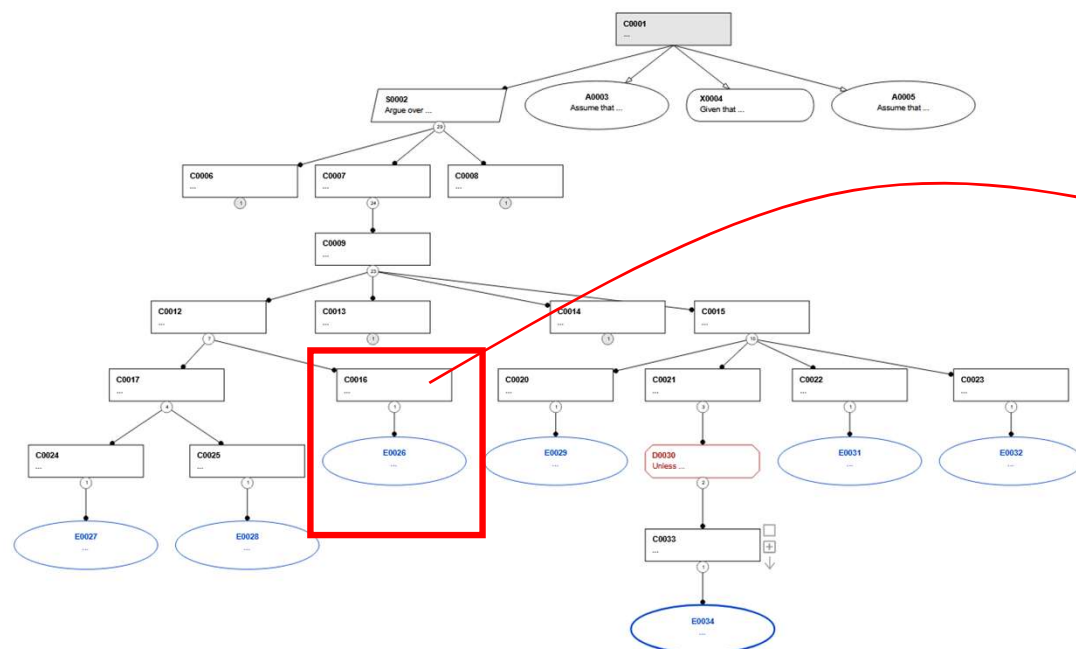
- Problem : Formal methods are used to produce verification evidence
 - That will need to be explained.
 - That can be combined with other kinds of evidence.

- Solution: A structured argument
 - That provides the necessary explanation from the top-level claim to the available evidence.
 - That clearly identifies the scope, assumptions and limitations.

Argument Structure

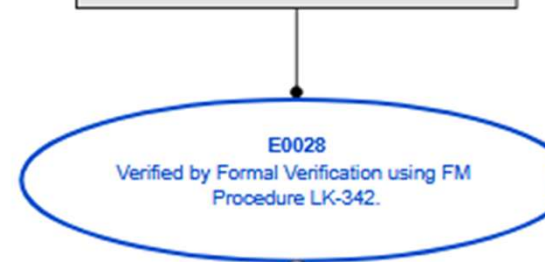




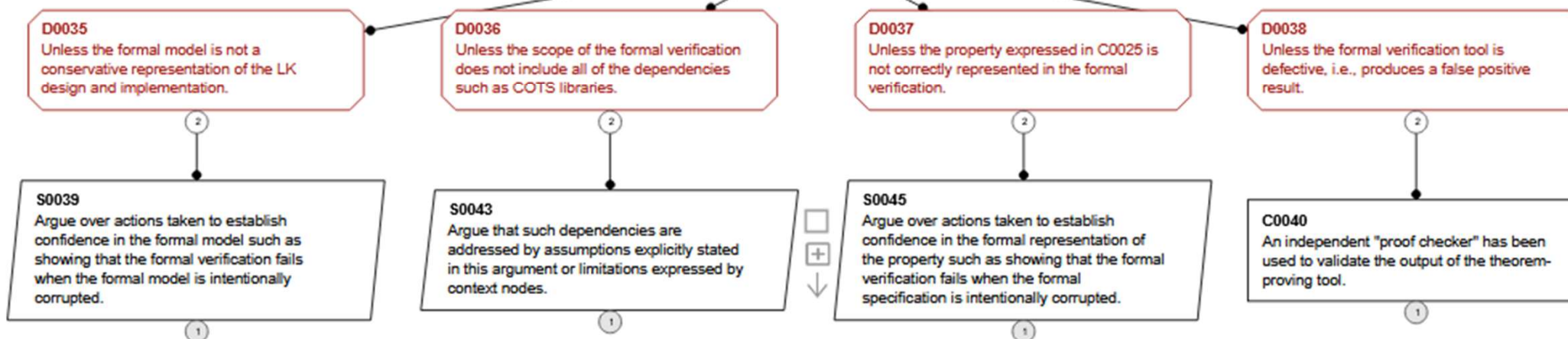
$$\forall v \forall o \forall h \forall p [(v < Lv_{max} \ \& \ o < LO_{max} \ \& \ h < CHA_{max}) \rightarrow (|p - LaneCenter| \leq 1 \text{ if } s = LKS(v, o, h))]$$


C0025
Given the current trajectory of the vehicle, if the current lateral velocity is less than LV_{max}, the current lateral offset if less than LO_{max} and the current relative heading angle is less than CHA_{max}, the automated driving system will request a change to the steering angle that will prevent lane departure if executed correctly.

Links to formal verification results



FM LK-342
executable
proof script



Change Management

1. Classify the changed in the system
2. Trace dependencies to claims and assumptions
3. Reassess the proof:
Locate affected models, lemmas and obligations
4. Require replayed proofs;
record failures and new limitations
5. Establish a new baseline

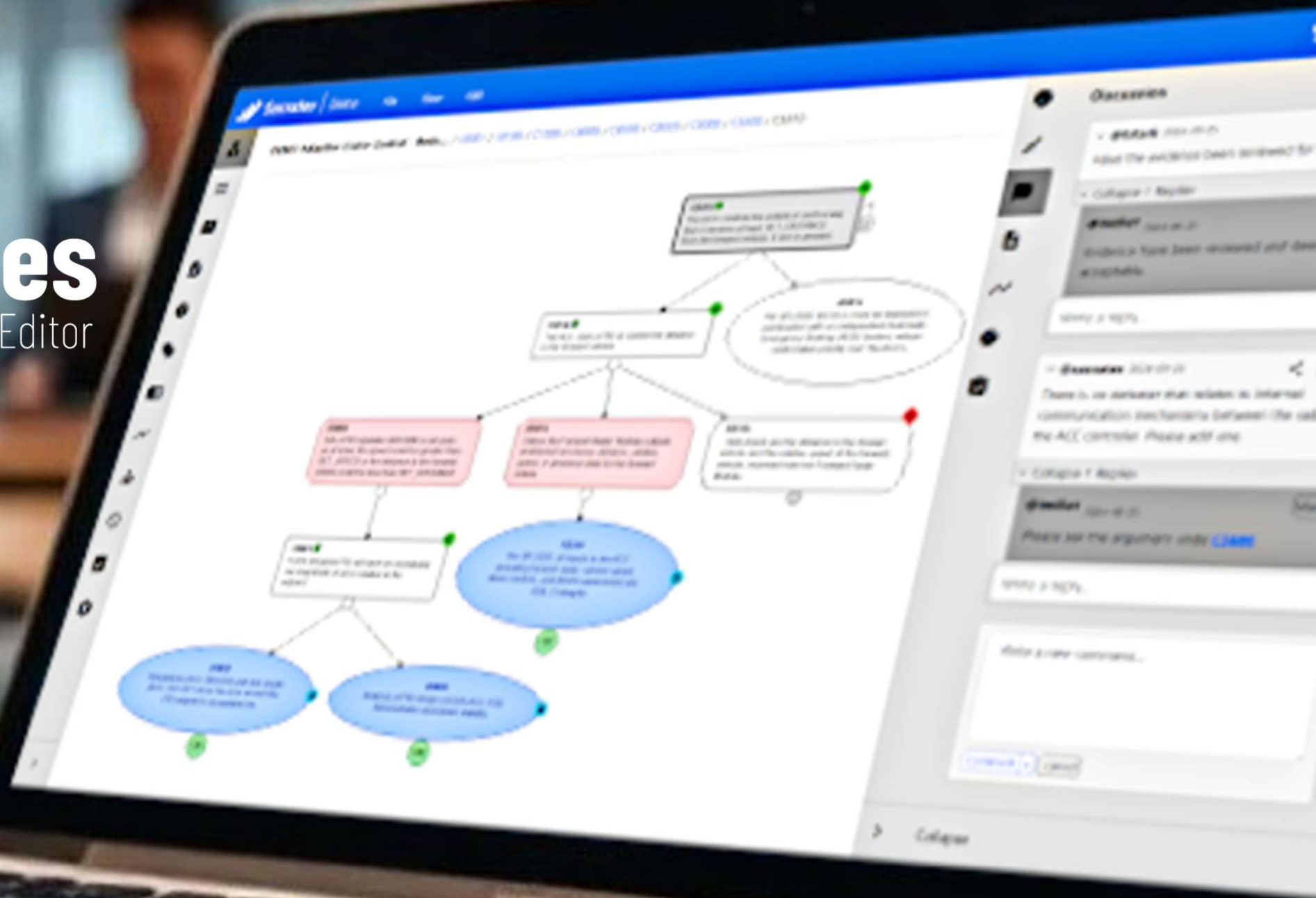


SPIs reveal broken assumptions

- Each SPI monitors an assumption or a limitation
- Can identify:
 - Configuration drift
 - Operating envelope excursions
 - ...
- Provides actionable evidence that the argument is incomplete

Maintaining confidence over the lifecycle

- Explain the evidence
- Manage change
- Monitor operations
- Reassess confidence



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AI-Enabled Monitoring of Space Robotics Telemetry

Applying Structured Argumentation to an AI/ML enabled system for monitoring telemetry communicated to a ground station by a robotic arm being used in human spaceflight.



Demonstration of a Safety Case Argument for the Gravity Ventilator

Illustrating how Eliminative Argumentation supports safety assurance for medical devices.



Generating Testable SOTIF Scenarios Using LLMs

Hieracillus: a Human-AI method that uses LLMs to expand and refine SOTIF scenario generation for safety analysis of advanced vehicles.



From Standard to System: ISO/PAS 8800 Assurance in Practice with Socrates

From design to deployment, ISO 8800 defines how to argue AI safety with logic, evidence and live metrics.



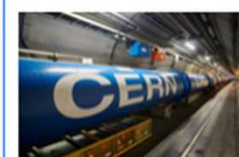
First of its Kind - The NRC Sees the Argument

First Instance of the use of an assurance case in the US NRC regulatory environment, using structured argumentation.



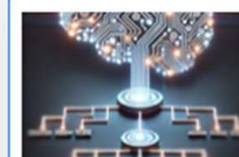
LLM & Structured Argumentation Leads to Trust

LLMs support safety-critical decisions by evaluating structured arguments, identifying gaps, surfacing doubts, and generating defeaters to challenge bias and strengthen trust.



Assurance Cases in the Large

See how CERN's Machine Protection System uses structured argumentation to mitigate risks in the world's largest particle accelerator, ensuring operational safety.



Generating Defeaters with Gen AI

Use Generative AI to help brainstorm defeaters for structured arguments.



Challenging Autonomy with Combinatorial Testing

With trillions upon trillions of possibilities, how could you identify a set of test cases for an autonomous system that is both manageable and adequate?



Cross-Sector Application of ISO/PAS 8800 for AI & ML Safety

Guidance on applying ISO/PAS 8800 across industries for AI/ML safety assurance.



Bridging the Gap Between IEC 61511 and Use of AI in Plant Safety

Bridging deterministic safety and AI uncertainty: a roadmap to reconcile IEC 61511 with the future of intelligent plant systems.



Uncovering Unsafe Feature Interaction in Vehicle Control using Generative AI

Harnessing LLMs and Digital Twins to uncover hidden ADAS risks faster, cheaper, and smarter.



Establishing Trust in Data for Critical Decision Making

Using structured argumentation to establish trust in in-line inspection (ILI) results used to evaluate the integrity of a pipeline.



Functional Safety Requirements for Artificial Intelligence and ML Systems

Discover how Metamorphic Relations offer a breakthrough in defining functional safety requirements for AI/ML, tackling the 'black-box' problem.



Confirmation Bias & Safety Management Systems

Discover how Eliminative Argumentation and AI enhance Safety Management Systems by reducing bias and providing real-time performance insights through KPI integration.



Safety Integrity Levels for Artificial Intelligence

Re-thinking how level-of-rigour approaches for conventional software are applied to AI-enabled systems.



Incremental Assurance Through Eliminative Argumentation

How can an assurance case represent how confidence in an argument changes over time?



Assurance Case Development as Data

Could AI learn how to guide a safety engineer in the development of a safety case?



A Fuzzy Logic Language for Trustworthy Confidence in Assurance Cases

Certus: Bringing clarity to assurance confidence with fuzzy logic and a language engineers can trust.



Not a "One Size Fits All" for AI in Safety Assurance

A call to tailor LLM evaluation methods to each assurance case use, risk profile, and role.



Balancing AI's Promise and Peril in Assurance

A practical framework for weighing the risks and rewards of applying LLMs in safety assurance cases.



Electric Over Water - Analysis of Hazards for an All-Electric Floatplane

Electric propulsion is reshaping aviation, but can safety keep pace with innovation? Explore how Eliminative Argumentation addresses the hazards of tomorrow's aircraft.



Utilizing an Assurance Case Argument to Drive Development

How can a safety case transcend being merely a final step in the safety process?



Adding Defeaters to Confidence Assessment

Take account of defeaters and doubt in confidence assessment methods, including those used in our Socrates product.



Patterns for Security Assurance Cases

Re-useable patterns, based on NIST 800-93, for creating security assurance arguments.



Safety for Cyber-Physical Systems

A practical approach to applying formal methods to guide the development of numerically intensive control systems.



Eliminative Argumentation for Arguing System Safety

Using the power of "doubt" to overcome confirmation bias and harden a safety assurance case.





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